

Unit - 2

Minimization of switching functions

→ In the previous chapter, we simplified switching expressions by means of algebraic manipulations.

→ It is not an algorithmic method and ineffective. Very tedious as no. of variables and terms increases.

→ $f(x_1, x_2, \dots, x_n)$ function given
 $g(x_1, x_2, \dots, x_n)$ simplified fn.

Which is equivalent to f and minimizes some cost criteria.

- This results into criteria based on the cost
- ① minimum no. of terms in sop or in pop
 - ② minimum no. of literals in sop or in pop
 - ③ minimum no. of appearance of literals

→ $f = xy + xz + x'y'$

↳ more literals.

$x(y+z) + x'y'$

↳ fewer literals.

→ $f = \underbrace{x'y'z' + x'y'z}_{x'z'} + \underbrace{x'y'z' + x'y'z}_{y'z'} + \underbrace{x'y'z' + x'y'z}_{yz} + \underbrace{x'y'z' + x'y'z}_{xz}$

$f = \Sigma m(0, 2, 3, 4, 5, 7)$

$f(x, y, z) = x'z' + xy' + yz \rightarrow \text{solution 1}$

$x'z' + y'z' + yz + xz \rightarrow \text{solution 2}$

$x'y' + y'z' + yz \rightarrow \text{solution 3}$

All the 3 solutions are irredundant, sol ① & sol ② are two minimal solutions.

∴ An irredundant expression is not necessarily minimal, nor is the minimal expression always unique.

→ To overcome this drawback the simplification method

- ① Map Method (Karnaugh method / K-map method)
- ② Tabulation method / Quine Mccluskey Method.

K-map:

- is a graphical representation that provides a systematic method for simplifying and manipulating boolean expressions.
- Map is a modified form of a truth table in which the arrangement of the combinations is convenient.
- The structure of the map changes with the no. of i/p variables of the function.
- Each n -variable map consists of 2^n cells (squares), representing all possible combinations of i/p variables.

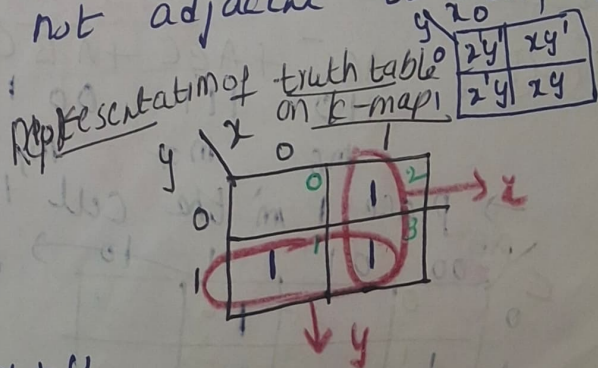
No. of Variables (n)	No. of cells in the map (2^n)
1	$2^1 = 2$ cells
2	$2^2 = 4$ cells
3	$2^3 = 8$ cells
4	$2^4 = 16$ cells
5	$2^5 = 32$ cells

- Beyond 5 variable it is a tedious task to represent the map.
- In the map, cells are adjacent to other, if they vary in only one variable position and all other remaining positions are same.
- The diagonal cells are not adjacent to each other.

Ex:- 2-Variable map

x	y	F
0	0	0
0	1	1
1	0	1
1	1	1

$$\therefore F = x + y$$



→ In a 3-variable map every cell is adjacent to other 3 cells.

xy \ xz	00	01	11	10
0	0	1	2	3
1	4	5	6	7

→ minterms.

xy \ xz	0	1
00	0	1
01	2	3
10	4	5
11	6	7

→ In a 4-variable map every cell is adjacent to other 4-cells.

yz \ wx	00	01	11	10
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

→ In the map we have 2^M cells, each cell is adjacent to M -cells of the collection, it's called a subcube, and a subcube is said to cover these cells.

→ Each subcube can be expressed by a product containing $n-m$ literals. where n = no. of variables in the fn.

xy \ xz	00	01	11	10
0	1	1		
1	1	1		

3-variable map $n=3$

4-cells subcube: i.e. $2^m = 4 \Rightarrow m=2$

∴ no. of literals in the product term
 $= n - m = 3 - 2 = 1$ literal.

Plotting canonical sop and pos on k-map.

Ex-① $y = ABC + ABC + \bar{A}\bar{B}C = y = \sum m(1, 6, 7)$

→ place 1 in the cell 1, 6 and 7

C \ AB	00	01	11	10
0	0	2	6	4
1	1	3	7	5

↑ $\bar{A}\bar{B}C$ ABC

Ex-② $y = (A + \bar{B} + C)(A + \bar{B} + \bar{C})(A + \bar{B} + C)(\bar{A} + \bar{B} + C)$

$y = \prod M(1, 2, 3, 6)$

C \ AB	00	01	11	10
0	0	2	6	4
1	0	3	7	5

Grouping of cells for simplification

→ Boolean expression is represented by canonical SOP and POS expression

→ We know that minterms (SOP) are marked by 1s and

Maxterms are marked by 0s.

→ Simplify logic function by using grouping of cells. Grouping means combining adjacent cells which contain 1s for

SOP (or) 0s for POS.

→ Grouping cells are selected in the form of powers of 2 i.e. ($2^0, 2^1, 2^2, \dots$). Take the common variables from the combined cells to get simplified expression.

xy \ z	00	01	11	10
0	1	1	0	0
1	0	1	0	1

→ $\bar{x}\bar{y}$

xy \ z	0	1
0	1	1
1	0	1

same value in both the cells.

different values.

$$f = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z}$$

$$f = \bar{x}\bar{z}(\bar{y} + y) = \bar{x}\bar{z}$$

Grouping of two adjacent cells removes one variable

xy \ z	00	01	11	10
0	1	1	0	0
1	0	1	0	1

→ $\bar{x}y$

xy \ z	00	01	11	10
0	0	0	0	0
1	1	1	1	1

→ $\bar{y}\bar{z}$

xy \ z	00	01	11	10
0	0	1	1	0
1	1	1	1	1

→ $\bar{z}y$

Overlapping pairs.

xy \ z	00	01	11	10
0	1	1	0	0
1	0	1	0	1

→ $\bar{x}\bar{z}$

Combination not required

$$f = y\bar{z} + \bar{x}\bar{z}$$

xy \ z	00	01	11	10
0	0	0	0	0
1	1	1	1	1

→ y

Rows
Columns
eliminate from
y

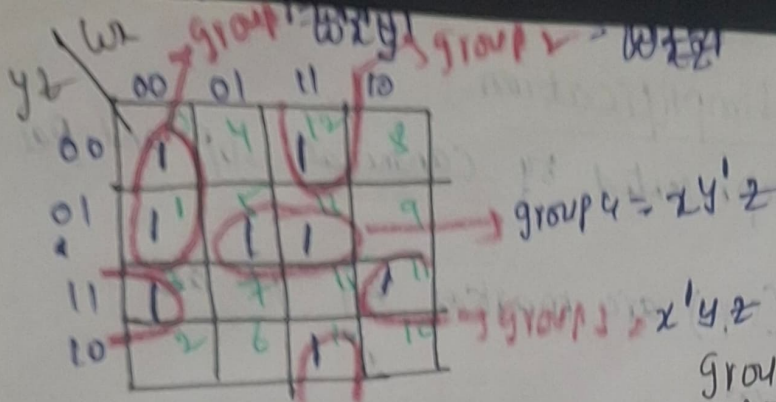
eliminate

Grouping of 4-adjacent cells remove 2-variables

Grouping of 8-adjacent cells remove 3-variables

xy \ z	00	01	11	10
0	1	1	1	1
1	1	1	1	1

→ $1 = 1$



group 2: $\overline{w}x'y'z$

Common: $w'z$

group 1: $w'z$

Common Variables: $w'z$

different values remove

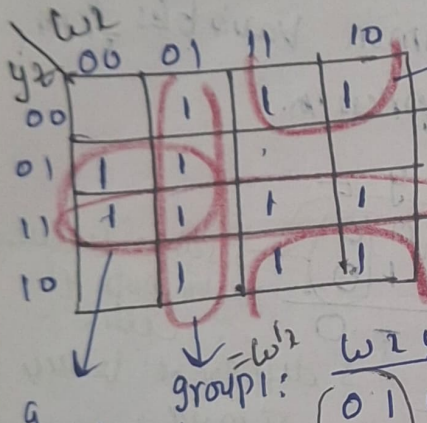
group 3: $(3, 11)$

Common Variables: $xy'z$

group 4: $(5, 13)$

Common Variables: $w'z$

Common Variables: $w'z$



Common Variables: $w'z$

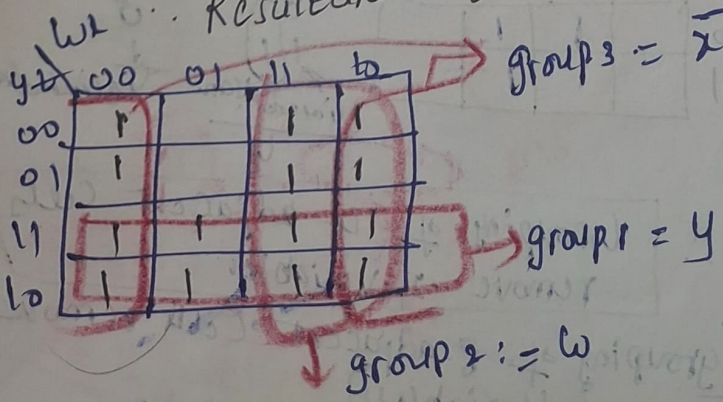
$$f = w'z + yz + w'z' + w'z$$

Common Variable: $w'z$

Grouping of Four 1's (quad) eliminates 2-Variables

that appear in both normal and complement form

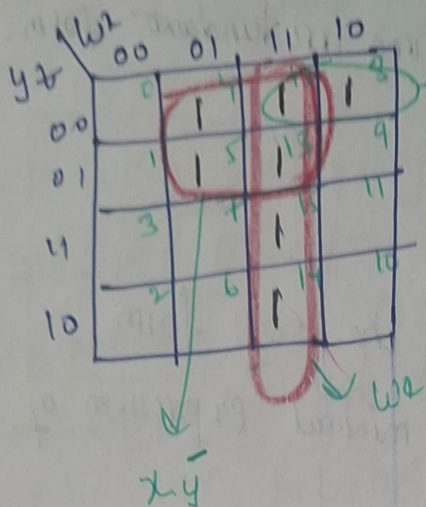
Resultant term contains only (n-2) Variables



grouping of eight 1's eliminates 3-Variables

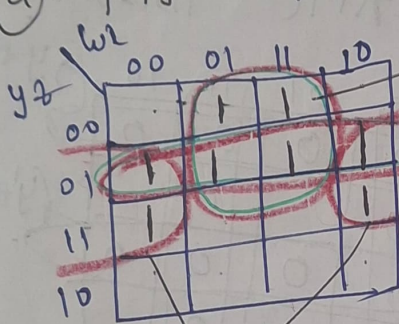
resultant product term contains (n-3) Variables

Ex: Simplify the following expression using map method.
 $f(w, x, y, z) = \sum m(4, 5, 8, 12, 13, 14, 15)$
 $f(w, x, y, z) = w\bar{x} + x\bar{y} + w\bar{y}z$ (Ans):



Gate problem:

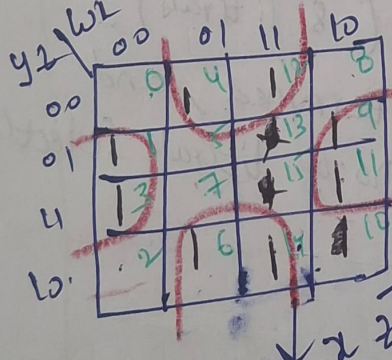
(a) Simplify the following fn $f(w, x, y, z) = \sum m(1, 3, 4, 5, 9, 11, 12, 13)$



$f = x\bar{y} + yz + z\bar{x}$
 can be avoided (already covered)

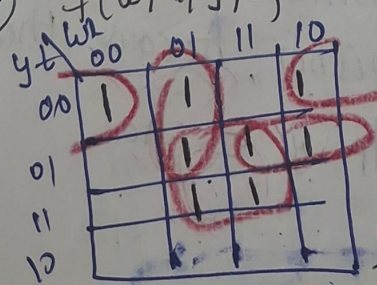
$$x\bar{y} + yz + z\bar{x}$$

(b) $f(w, x, y, z) = \sum m(1, 3, 4, 6, 9, 11, 12, 14)$

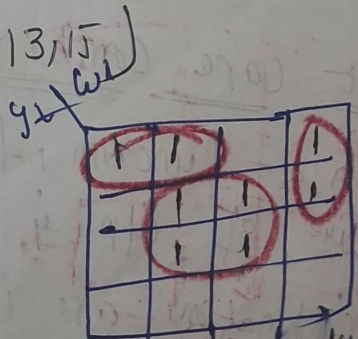


$$f = x\bar{z} + xz = x \oplus z$$

(3) $f(w, x, y, z) = \sum m(0, 4, 5, 7, 8, 9, 13, 15)$



$f = w\bar{y}z' + w'zy' + z$
 is an irreducible expression.



$$f = w\bar{y}z' + w'zy' + z$$

unique minimal expression

④ $f(w, x, y, z) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

wz	00	01	11	10
yz				
00			1	
01	1	1	1	
11		1	1	1
10		1		

$$f = wzy' + wyz + w'xy + w'yz$$

It has only one irredundant form.

Minimal Product of Sums

→ It is the dual procedure of the SOP form.

→ no. of literals required in the minimal expression of both the forms are not same.

6) $f(w, x, y, z) = \sum (5, 6, 9, 10)$

wz	00	01	11	10
yz				
00				
01		1		1
11	3	7		
10	2	1		1

$$f = w'xy'z + w'xyz' + wx'y'z + wx'yz'$$

(16 literals)

wz	00	01	11	10
yz				
00	0	0	0	0
01	0		0	
11	0	0	0	0
10	0		0	

$$f = (w+x)(y+z)(y'+z')(w'+x')$$

(8 literals)

Both the forms must be determined, and with the one with a smaller no. of literals selected.

Don't care combinations:

→ In a function, for certain combinations the value of o/p is unspecified. Either because these i/p combinations are invalid (i) o/p is of no importance.

→ combinations for which the value of the function is not specified are called "Don't Care Combinations".

→ It is denoted by ϕ (or d) (or x)

→ ϕ means either 0 or 1.

→ When employing the map of an incompletely specified function, we assign the value 1 to selected don't care combinations and value 0 to others, to increase the size of the subcubes whenever possible.

→ No subcubes containing only don't care cells may be formed.

Ex:- Design a code converter which converts BCD into ^{message} Excess-3 Code.

Decimal number	BCD-inputs w x y z	f ₄ f ₃ f ₂ f ₁ Excess-3 - outputs	Gray Code
0	0 0 0 0	0 0 1 1	0 0 0 0
1	0 0 0 1	0 1 0 0	0 0 0 1
2	0 0 1 0	0 1 0 1	0 0 1 1
3	0 0 1 1	0 1 1 0	0 0 1 0
4	0 1 0 0	0 1 1 1	0 1 1 0
5	0 1 0 1	1 0 0 0	0 1 1 1
6	0 1 1 0	1 0 0 1	0 1 0 1
7	0 1 1 1	1 0 1 0	0 1 0 0
8	1 0 0 0	1 0 1 1	1 1 0 0
9	1 0 0 1	1 1 0 0	1 1 0 1

if combinations 10 to 15 will never occur, may be regarded as don't care combinations.

$$f_1 = \sum m(0, 2, 4, 6, 8) + \sum \phi(10, 11, 12, 13, 14, 15)$$

$$f_2 = \sum m(0, 3, 4, 7, 8) + \sum \phi(10, 11, 12, 13, 14, 15)$$

$$f_3 = \sum m(1, 2, 3, 4, 9) + \sum \phi(10, 11, 12, 13, 14, 15)$$

$$f_4 = \sum m(5, 6, 7, 8, 9) + \sum \phi(10, 11, 12, 13, 14, 15)$$

f_1

yz	$w=00$	01	11	10
00	1	1	0	1
01	1	1	0	1
11	1	1	0	1
10	1	1	0	1

$f_1 = z'$

f_2

yz	$w=00$	01	11	10
00	1	1	X	1
01			X	
11	1	1	X	X
10			X	X

$f_2 = y'z' + yz$

f_3

yz	$w=00$	01	11	10
00		1	X	
01	1		X	1
11	1		X	X
10	1		X	X

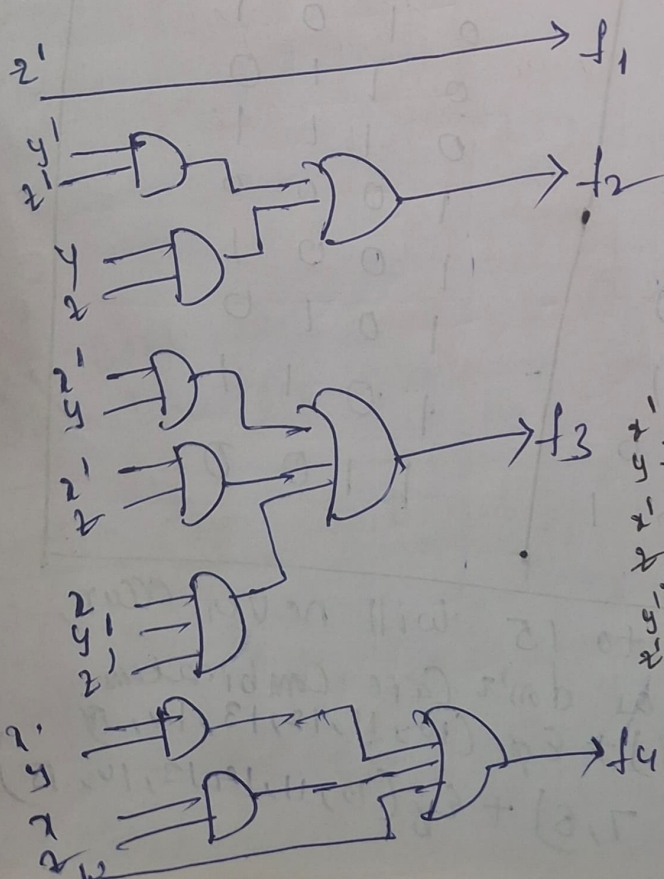
f_4

yz	$w=00$	01	11	10
00			X	1
01		1	X	1
11		1	X	X
10		1	X	X

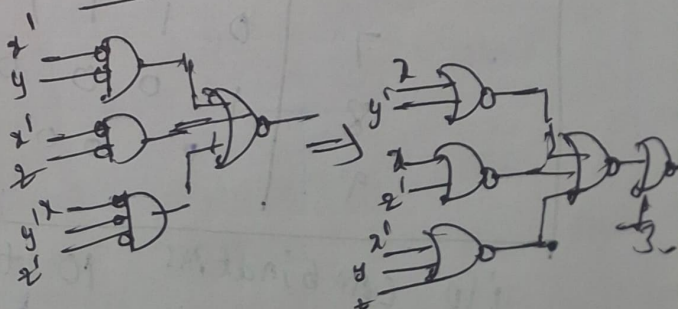
$$f_3 = z'y + z'z + zy'z'$$

$$f_4 = w + zy + z'z$$

NAND logic

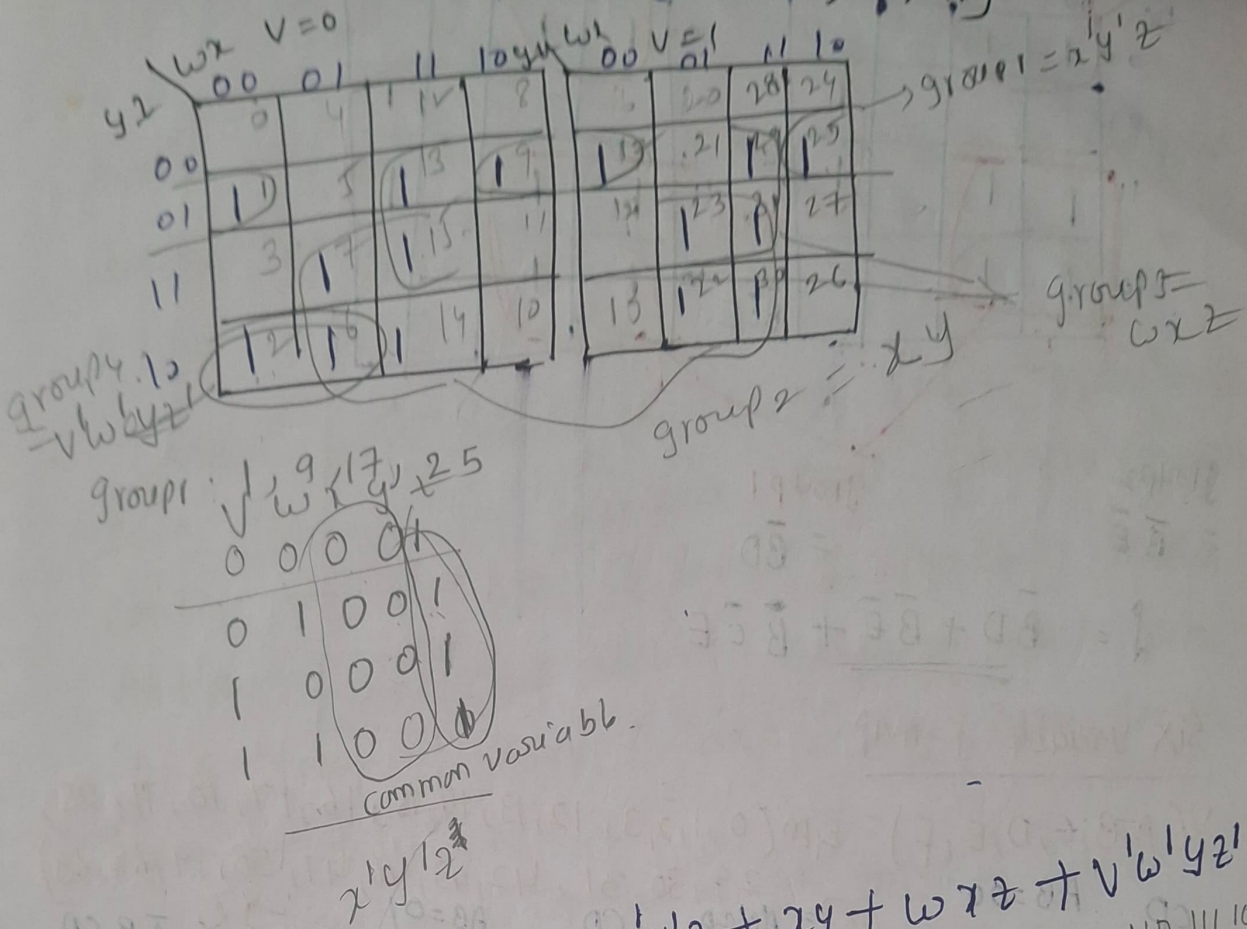


NOR logic



$$y'z' + yz = (y'z')' (yz)' = (y+z)(y'+z') = yz' + y'z + yz + y'z' = yz' + y'z + 1 = 1$$

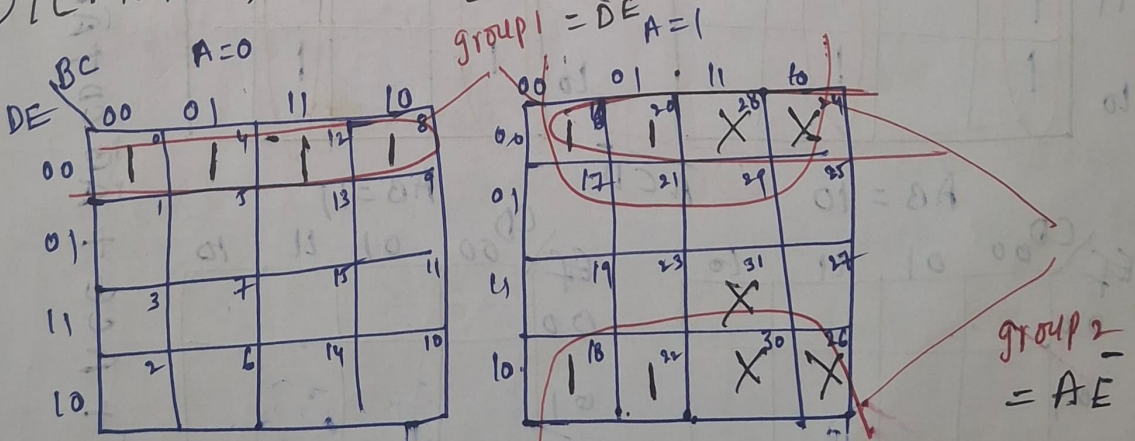
With the aid of map minimize the function
 Ex:- $f(V, w, x, y, z) = \sum m(1, 2, 6, 7, 9, 13, 14, 15, 17, 22, 23, 25, 29, 30, 31)$



$$f(V, w, x, y, z) = x'y'z + xy + wxz + V'w'yz'$$

Ex:- simplify the following using k-map method

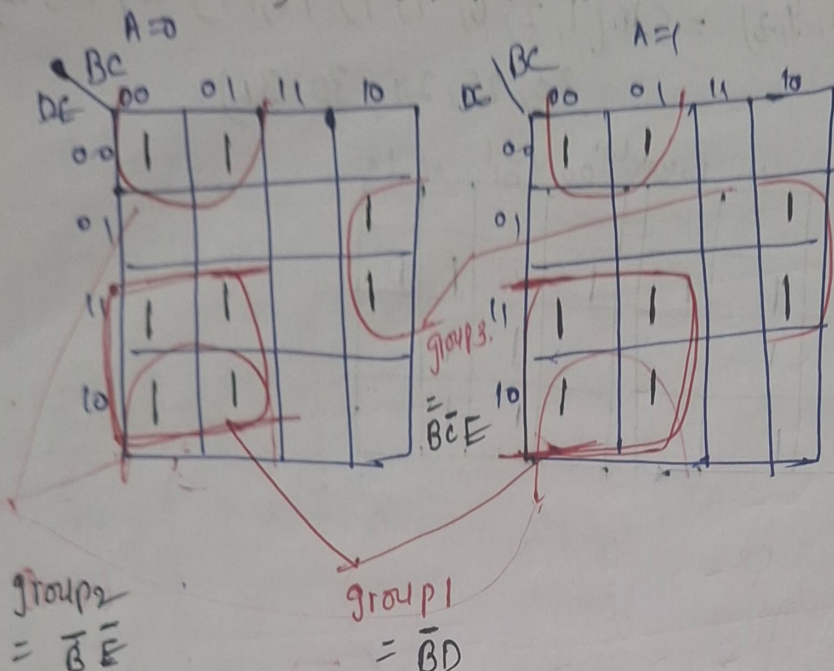
(a) $Y(A, B, C, D, E) = \sum m(10, 4, 8, 12, 16, 18, 20, 22) + \sum d(24, 26, 28, 30, 31)$



$$Y = \bar{D}\bar{E} + AE$$

$$= \bar{E}(A + \bar{D})$$

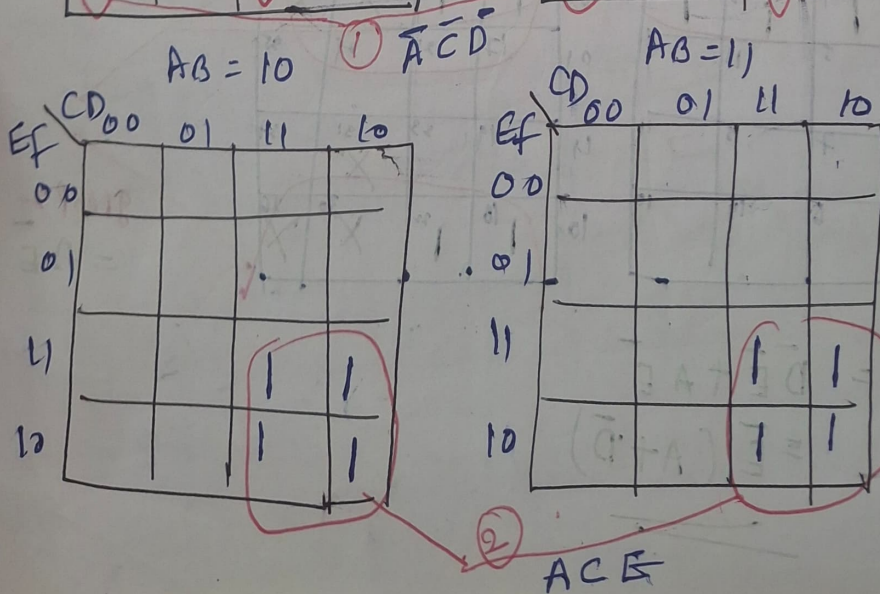
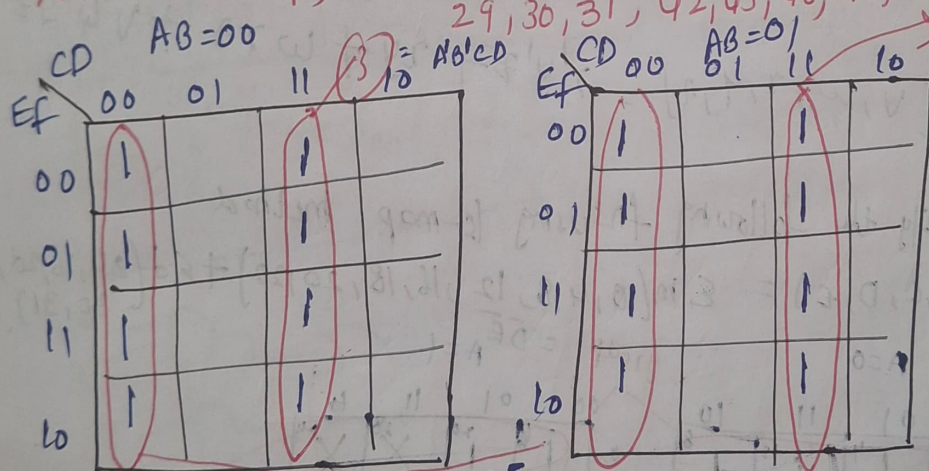
⑤ $Y(A,B,C,D,E) = \sum m(0,2,3,4,6,7,9,11,16,18,19,20,22,23,25,27)$



$$f = \bar{B}D + \bar{B}\bar{E} + \bar{B}\bar{C}E$$

Six variable K-map

$Y(A,B,C,D,E,F) = \sum m(0,1,2,3, 12,13,14,15, 16,17,18,19, 28, 29,30,31, 42,43,46,47, 58,59,62,63)$



$$f = \bar{A}\bar{C}\bar{D} + A\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D}$$

$$ACE$$

Quine-Mccluskey (Tabulation Method)

1. Arrange all minterms in groups, such that all the terms in the same group have the same no. of 1's in their binary representation. Start with the least no. of 1's and continue with groups of increasing no. of 1's. No. of 1's in a term is called the "index" of that term.
2. Compare every term of the lowest index group with each term in the successive group. Whenever possible, combine the two terms by means of combining theorem $Aa + Aa' = A$. Repeat this by comparing each term in a group of index i with every term in the group of index $i+1$.

Combined term consists of the original fixed representation, with different digit replaced by dash (-). A check mark (✓) is placed next to every term which has been combined with at least one term.

3. The terms generated in step 2 are now compared in the same fashion. A new term is generated by combining two terms which differ by only a single position and whose dashes are in the same position.

This process continues until no further combinations are possible. The remaining unchecked terms constitute the set of prime implicants of the function.

$E(w, 2, 4, 12) = E(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$

		Step (1)			
index (i)		w	x	y	z
i=0		0	0	0	0 ✓
i=1	1	0	0	0	1 ✓
	2	0	0	1	0 ✓
	8	1	0	0	0 ✓
i=2	5	0	1	0	1 ✓
	9	1	0	0	1 ✓
	10	1	0	1	0 ✓
i=3	7	0	1	1	1 ✓
	13	1	1	0	1 ✓
i=4	15	1	1	1	1 ✓

		Step (2)			
(0,1)		0	0	0	✓
(0,2)		0	0	0	✓
(0,8)		-	0	0	0 ✓
(1,5)		0	-	0	1 ✓
(1,9)		-	0	0	1 ✓
(1,10)		-	0	1	0 ✓
(2,10)		1	0	0	- ✓
(8,9)		1	0	-	0 ✓
(8,10)		1	0	-	0 ✓
(5,7)		0	1	-	1 ✓
(5,13)		-	1	0	1 ✓
(9,13)		1	-	0	1 ✓
(7,15)		-	1	1	1 ✓
(13,15)		1	1	-	1 ✓

		Step (3)			
(0,1,8,9)		-	0	0	- A
(0,2,8,10)		-	0	-	0 B
(1,5,9,13)		1	-	0	1 C
(5,7,13,15)		-	1	-	1 D
(0,8)		-	-	-	-
(2,10)		-	-	-	-
(4,12)		-	-	-	-
(2,8)		-	-	-	-

Note: In decimal method only
 (9,7) x → we can't compare because 9 is lower index higher decimal value and 7 is the higher index lower decimal value.

A, B, C and D are prime implicants. Next form prime implicant chart to get the simplified boolean expression.

		minterms									
Prime implicants		0	1	2	5	7	8	9	10	13	15
Essential row	A = x'y'	x	x				x	x			
	B = x'z' ✓	x		x			x		x		
Essential row	C = y'z		x		x				x	x	
	D = xz ✓				x	x				x	x

→ Select minimal subset of prime implicants such that each column contain at least one x that corresponding PI is essential.

→ once all the essential prime implicants has been selected, all the minterms it covers are checked off.
 → B and D cover all the minterms of function except 1 and 9. These minterms can be covered by either prime implicant A (or) C (both are expressed with same no. of literals).

∴ minimal expression

$$f = B + D + A = x'z' + xz + x'y'$$

(or)
 $B + D + C = x'z' + xz + y'z$

Decimal Representation

$$f(V, W, X, Y, Z) = \Sigma(13, 15, 17, 18, 19, 20, 21, 23, 25, 27, 29, 31) +$$

(0,1) → $\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & - & \end{array}$ 0,1 → 1)

$$\Sigma(1, 2, 12, 24)$$

→ further simplified by adopting decimal code.
 for minterms, rather than their binary representation.

→ two minterms can be combined only if they differ by a power of 2, i.e. only if the diff of their decimal code is 2^i .

→ The combined term consists of same literals with the exception of the variable whose weight is 2^i , which is deleted.

→ two terms whose code differ by a power 2 but which have the same index can't be combined, b/c if a term with a smaller index has a higher decimal value than another term whose index is higher, can't be combined, although they may differ by a power of 2. (Necessary but not sufficient condition)

$$f = E(13, 15, 17, 18, 19, 20, 21, 23, 25, 27, 29, 31) + \Sigma \phi(1, 2, 12, 24)$$

1 ✓	17, 19, 21, 23 (2, 4) ✓
2 ✓	17, 19, 25, 27 (2, 8) ✓
	17, 21, 25, 29 (4, 8) ✓
i=2 12	13, 15, 29, 31 (2, 16) B ✓
17 ✓	19, 23, 27, 31 (4, 8) ✓
18 ✓	21, 23, 29, 31 (2, 8) ✓
20 ✓	25, 27, 29, 31 (2, 4) ✓
24 ✓	
i=3 13 ✓	17, 19, 21, 23, 25, 27, 29, 31 (2, 48) A ✓
19 ✓	
21 ✓	
25 ✓	
i=4 15 ✓	
23 ✓	
27 ✓	
29 ✓	
i=5 31 ✓	

(17, 13) X → lower index higher decimal and 13 is higher index
lower decimal ∴ Can't be Combined.

27, 31 (4) ✓
29, 31 (2) ✓
13 15 17 18 19 20 21 23 25 27 29 31

A ✓		X	X	X	X	X	X	X	X
B ✓	X	X						X	X
C						X			
D ✓				X	X				
E			X	X					
F	X								
G			X						
H		X							

→ Essential prime implicants are A, B and D. They cover all the minterms of the given function with the exception of 18.

→ 18 can be covered by either PI $E = \overline{v} \overline{w} \overline{x} y z$

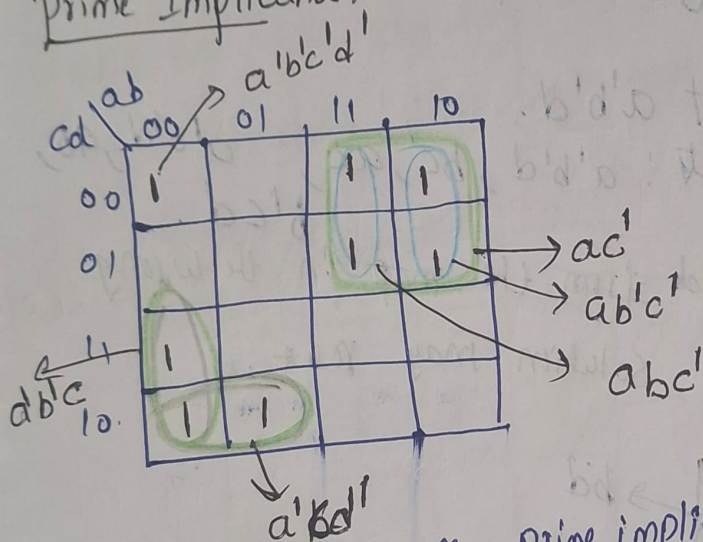
$$f = A + B + D + E = \underline{\underline{vz + wz + vw'zy' + vw'zy}}$$

(or)

$$A + B + D + G = \underline{\underline{vz + wz + vw'zy' + w'zyz}}$$

Determination of minimal Expressions Using Essential

Prime Implicants:



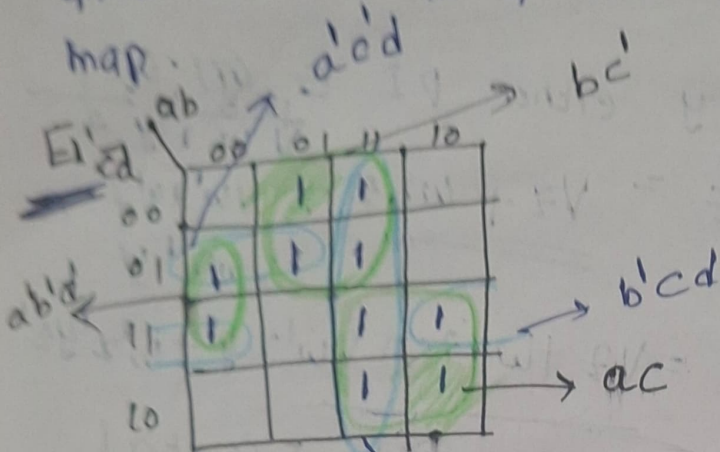
→ $a'cd'$, $a'b'c'$ are prime implicants because they can't be combined with other terms to eliminate a variable.

→ $a'b'c'd'$ is not a prime implicant because it can be combined with $a'b'cd'$ (or) $ab'c'd'$.

→ Neither $ab'c'$ nor abc' is a PI because they can be combined with ac' .

∴ PIs can be obtained from K-map. A single 1 on a map represents a PI if it is not adjacent to any other 1. Two adjacent 1s on map form a PI if they are not contained in a group of four. Four adjacent 1s form a PI if they are not combined in a group of eight, etc.

→ In order to find minimal SOP from a map, we must find minimum no. of P.I. which cover all the 1's on the map.

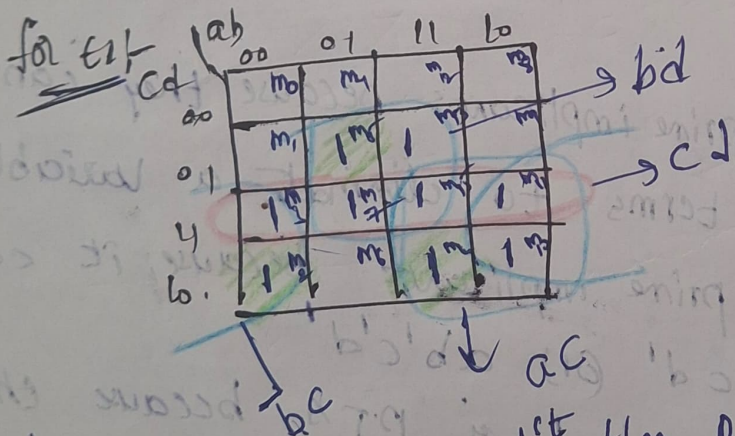


→ $ac, bc', a'b'd$ cover all the 1's on the map and minimum solution is the sum of these P.I.

$$\therefore F = ac + bc' + a'b'd.$$

All the prime implicants: $a'b'd, bc', ac, a'dc'd, ab, b'cd$.

→ If P.I. are selected from the map in the wrong order, a nonminimum solution may result.



→ if cd is chosen 1st then $BD, B'C, AC$ are needed to cover remaining 1's. and solution contains 4 terms.
→ If bd, bc, ac are chosen 1st then cd is not needed.

→ m_2 is covered by only $B'C$, but m_3 is covered by both $B'C$ and CD .

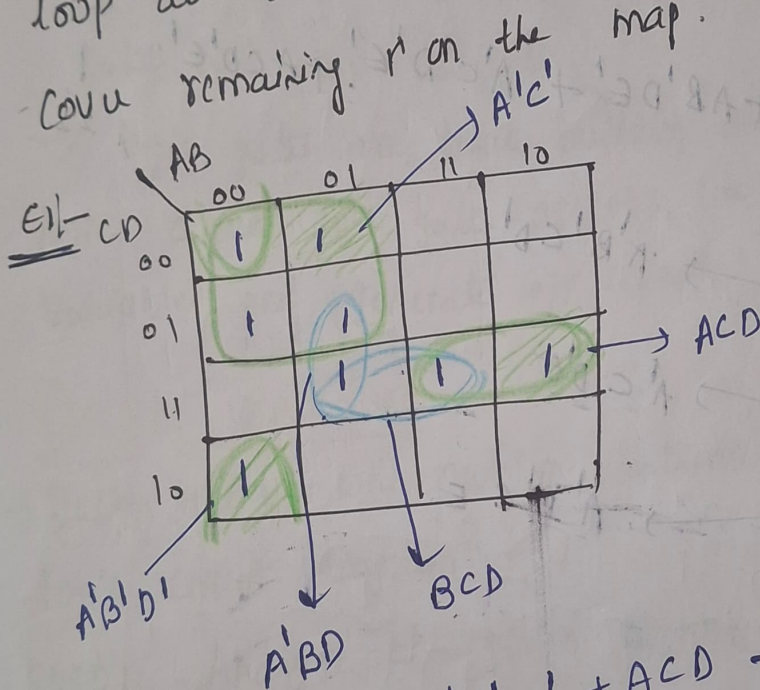
→ If a minterm is covered by only one prime implicant, that PI is said to be essential, and it must be included in the minimal sop.

→ m_5 is covered by BD, so BD is essential.

→ m_4 is covered by only AC, so AC is essential.

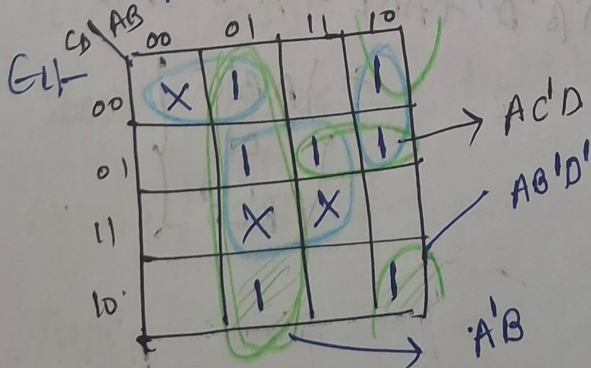
→ All essential PI, covers all the 1s on the map, and nonessential PI is not needed.

∴ In order to find minimal sops, we should loop all the essential PI. Next find PI which cover remaining 1s on the map.



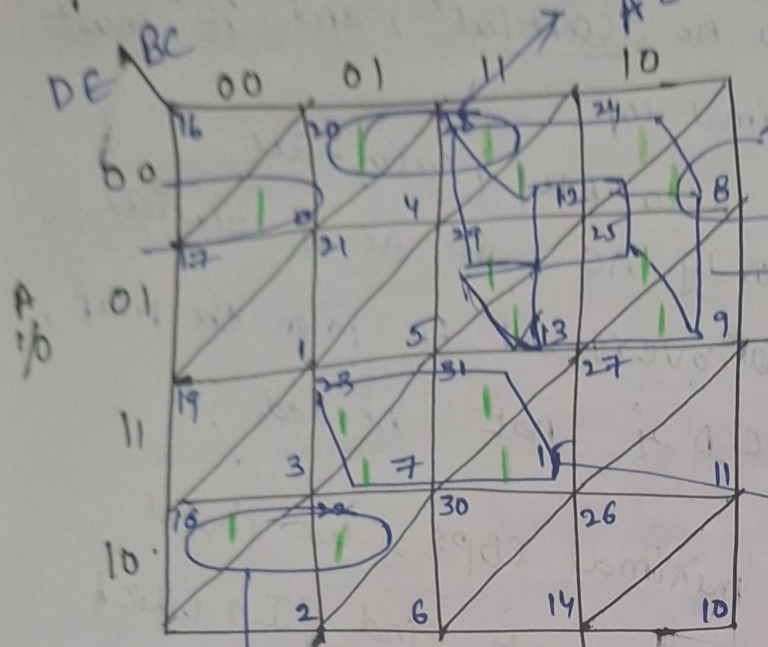
$$f = A'C' + \underbrace{A'B'D' + A'BD + BCD}_{\text{essential PIs}} + ACD$$

(or) $A'B'D$



$$f = A'B + \underbrace{A'B'D + A'B'D'}_{\text{essential}} + A'C'D$$

Ex- 5-Variable k-map



$$(0, 4, 16, 20) \rightarrow A'B'DE'$$

$$\rightarrow BD'$$

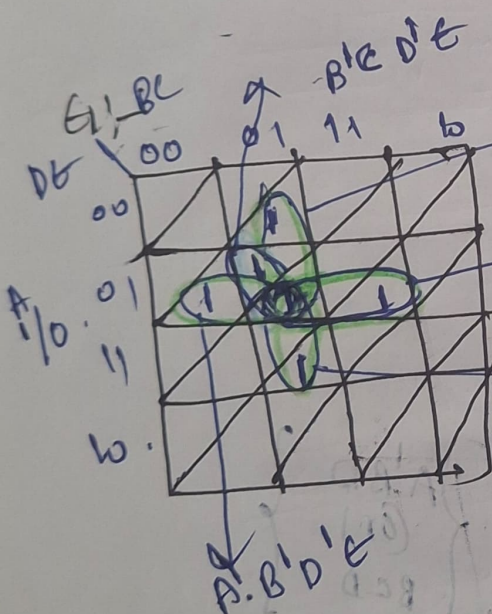
$$\rightarrow CDE$$

$$AB'DE'$$

$$f = BD' + CDE + AB'DE' + A'C'D'E' + ACD'E'$$

$$\begin{matrix} 20 & 10100 \\ 4 & 00100 \\ 12 & 11000 \end{matrix}$$

$$\begin{matrix} 16 & 20 & 28 \\ & 1 & 1 \\ 21 & & \\ & 1 & \\ & & 22 \end{matrix}$$



$$\rightarrow A'B'CD'$$

$$\rightarrow A'C'D'E'$$

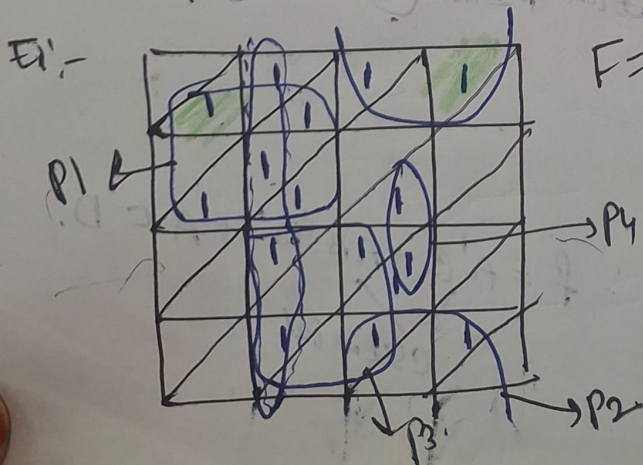
$$\rightarrow A'B'CE$$

$$A'B'D'E'$$

$$P_1 \quad P_2 \quad P_3 \quad P_4$$

$$F = A'B'D' + ABE' + ACD + A'BCE$$

$$+ \left. \begin{matrix} ABC \\ \text{or} \\ B'CD' \end{matrix} \right\}$$



Tabulation procedure

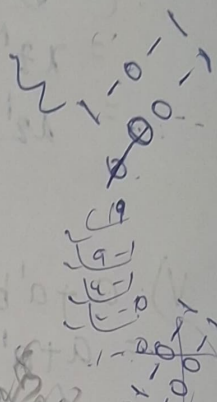
pg: 109

a) $f(w, x, y, z) = \sum (1, 5, 6, 12, 13, 14) + \prod (2, 4)$

i=0	1	0	0	0	1	✓
	2	0	0	1	0	✓
	4	0	1	0	0	✓
i=2	5	0	1	0	1	✓
	6	0	1	1	0	✓
	12	1	1	0	0	✓
i=3	13	1	1	0	1	✓
	14	1	1	1	0	✓

(1,5)	0	-	0	1	✓
(2,6)	0	-	1	0	✓
(4,5)	0	1	0	-	✓
(4,6)	0	1	-	0	✓
(4,12)	-	1	0	0	✓
(5,13)	-	1	0	1	✓
(6,14)	-	1	1	0	✓
(12,13)	1	1	0	-	✓
(2,14)	1	-	0	✓	

$(4,6,12,14) - 1 - 0$ (B)
 $(1,5,12,13) - 10 -$ (A)



A, B, D are essential PI

EPI

$f = A + B + D$
 $= xy' + xz + w'y'z$

A		x		x	x
B			x	x	x
C			x		
D	x	x			

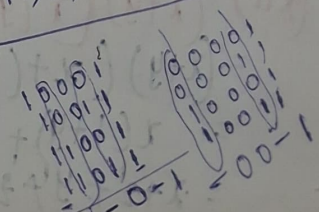
b) $f(w, x, y, z) = \sum (0, 1, 3, 8, 9, 13, 14, 15, 16, 17, 19, 24, 25, 27, 31)$

i=0	0	0	0	0	0	0	✓
i=1	1	0	0	0	0	1	✓
	8	0	1	0	0	0	✓
	16	1	0	0	0	0	✓
i=2	3	0	0	0	1	1	✓
	7	0	0	0	1	0	✓
	15	1	1	0	0	0	✓
	24	1	1	0	0	0	✓
i=3	13	0	1	0	1	1	✓
	14	0	1	1	1	0	✓
	19	1	0	0	1	1	✓
	25	1	1	0	0	1	✓
i=4	16	0	1	1	1	1	✓
	27	1	1	0	1	1	✓
i=5	31	1	1	1	1	1	✓

0,1 (1) ✓	1,3 (2) ✓	1,7 (16) ✓	8,9 (1) ✓	5,24 (16) ✓	16,17 (1) ✓	16,24 (8) ✓	3,19 (16) ✓	9,13 (4) ✓	9,25 (16) ✓	17,19 (2) ✓	24,25 (1) ✓	(1,25) (8) ✓
0,8 (8) ✓	1,9 (8) ✓	8,9 (1) ✓	5,24 (16) ✓	16,17 (1) ✓	16,24 (8) ✓	3,19 (16) ✓	9,13 (4) ✓	9,25 (16) ✓	17,19 (2) ✓	24,25 (1) ✓	(1,25) (8) ✓	
0,16 (16) ✓	1,17 (16) ✓	8,9 (1) ✓	5,24 (16) ✓	16,17 (1) ✓	16,24 (8) ✓	3,19 (16) ✓	9,13 (4) ✓	9,25 (16) ✓	17,19 (2) ✓	24,25 (1) ✓	(1,25) (8) ✓	

1, 3, 8, 9, 13, 14, 15, 16, 17, 19, 24, 25, 27, 31

(13,15) (2) ✓	14,15 (1) ✓	19,27 (8) ✓	25,27 (2) ✓	15,31 (16) ✓	27,31 (4) ✓	1,3,17,19 (2,16) ✓	1,17,19,25 (16,8) ✓	8,9,24,25 (1,16) ✓	16,17,24,25 (1,8) ✓	17,19,27,27 (2,8) ✓
(13,15) (2) ✓	14,15 (1) ✓	19,27 (8) ✓	25,27 (2) ✓	15,31 (16) ✓	27,31 (4) ✓	1,3,17,19 (2,16) ✓	1,17,19,25 (16,8) ✓	8,9,24,25 (1,16) ✓	16,17,24,25 (1,8) ✓	17,19,27,27 (2,8) ✓
(13,15) (2) ✓	14,15 (1) ✓	19,27 (8) ✓	25,27 (2) ✓	15,31 (16) ✓	27,31 (4) ✓	1,3,17,19 (2,16) ✓	1,17,19,25 (16,8) ✓	8,9,24,25 (1,16) ✓	16,17,24,25 (1,8) ✓	17,19,27,27 (2,8) ✓



Step 4

PI chart

Step 4

PI chart

	0,1,8,9	16,17,24,25	(1,8,16)	(A)	27	31					
	2	13	14	15	16	17	19	24	25		
A	X	X									
B											
C											
D											
E											
F											
G											
H											
I											

A, C, G are EPI

A, C, G are covering all the minterms except 13, 27, 31

27, 31 → D
13, 15 → H/I

Reduced PI chart

	13	27	31
B		X	
D		X	X
E			X
F		X	
H	X		
I	X		

← Row dominance

$$A + C + G + D + \begin{Bmatrix} H \\ I \end{Bmatrix}$$

4.7

$1 \times 0 + 0 \times 1 = 100$
 $f_1 = x_2' + x_1$
 $f_2 = x_1 x_2$
 $f = f_1 + f_2$
 $f = (0, 4, 9, 10, 11, 12)$

$f_1 = (1, 3, 4, 6, 9, 11, 12, 14)$

$f_2 = (4, 9, 11, 12) + \emptyset(1, 3, 6, 14)$

$f_3 = (0, 10)$

$0, 4, 9, 10, 11, 12$

$0 \ 2 \ 3 \ 2$
 $0 \ 1 \ 0 \ 0$
 $0 \ 1 \ 1 \ 0$
 $0 \ 1 \ 1 \ 0$
 $1 \ 1 \ 1 \ 1$

$X \ 0 \ 0 \ 1$
 $0 \ 0 \ 0 \ 1 \rightarrow 1$
 $0 \ 0 \ 1 \ 1 \rightarrow 3$
 $1 \ 0 \ 0 \ 1 \rightarrow 9$
 $1 \ 0 \ 1 \ 1 \rightarrow 11$

	step 1	step 2
i=1	1 ✓	1, 3 (2)
	4 ✓	1, 5 (4)
2	3 ✓	4, 5 (1)
	5 ✓	4, 6 (2)
	6 ✓	4, 12 (8)
	10 ✓	4, 20 (16)
	12 ✓	3, 7 (4)
	18 ✓	3, 11 (8)
3	20 ✓	3, 19 (16)
	7 ✓	5, 7 (2)
	11 ✓	5, 13 (8)
	13 ✓	5, 21 (16)
	14 ✓	6, 7 (1)
4	19 ✓	6, 14 (8)
	21 ✓	
	22 ✓	
	25 ✓	
	15 ✓	
	23 ✓	
	26 ✓	
	27 ✓	